PROJECT: UNDERSTANDING SUBSCRIPTION BEHAVIORS

A SaaS company seeks to uncover what drives its clients to renew subscriptions. They've collected data on client details, subscription records, and economic indicators and would like to connect them to better understand its clients' behavior.

They've tasked you with analyzing these datasets to identify the key factors influencing clients' decisions to renew their subscriptions.

Your analysis will provide them with insights into which customers are renewing their products and the reasons behind their renewals. The company can leverage these insights to make informed decisions to increase renewal rates and improve customer loyalty, helping them stay competitive and ensure long-term growth.

The Data

The company have provided you with three datasets for your analysis. A summary of each data is provided below.

`client_details.csv`

Column	Description
`client_id`	Unique identifier for each client.
`company_size`	Size of the company (Small, Medium, Large).
`industry`	Industry to which the client belongs (Fintech, Gaming, Crypto, AI, E-commerce).
`location`	Location of the client (New York, New Jersey, Pennsylvania, Massachusetts, Connecticut).

`subscription_records.csv`

Column	Description
`client_id`	Unique identifier for each client.
`subscription_type`	Type of subscription (Yearly, Monthly).
`start_date`	Start date of the subscription - YYYY-MM-DD.
`end_date`	End date of the subscription - YYYY-MM-DD.
`renewed`	Indicates whether the subscription was renewed (True, False).

`economic_indicators.csv`

Column	Description
`start_date`	Start date of the economic indicator (Quarterly) - YYYY-MM-DD.
`end_date`	End date of the economic indicator (Quarterly) - YYYY-MM-DD.
`inflation_rate`	Inflation rate in the period.
`gdp_growth_rate`	Gross Domestic Product (GDP) growth rate in the period.

```
# Re-run this cell
# Import required libraries
import pandas as pd

# Import data
client_details = pd.read_csv('data/client_details.csv')
subscription_records = pd.read_csv('data/subscription_records.csv', parse_dates =
['start_date','end_date'])
economic_indicators = pd.read_csv('data/economic_indicators.csv', parse_dates =
['start_date','end_date'])
```

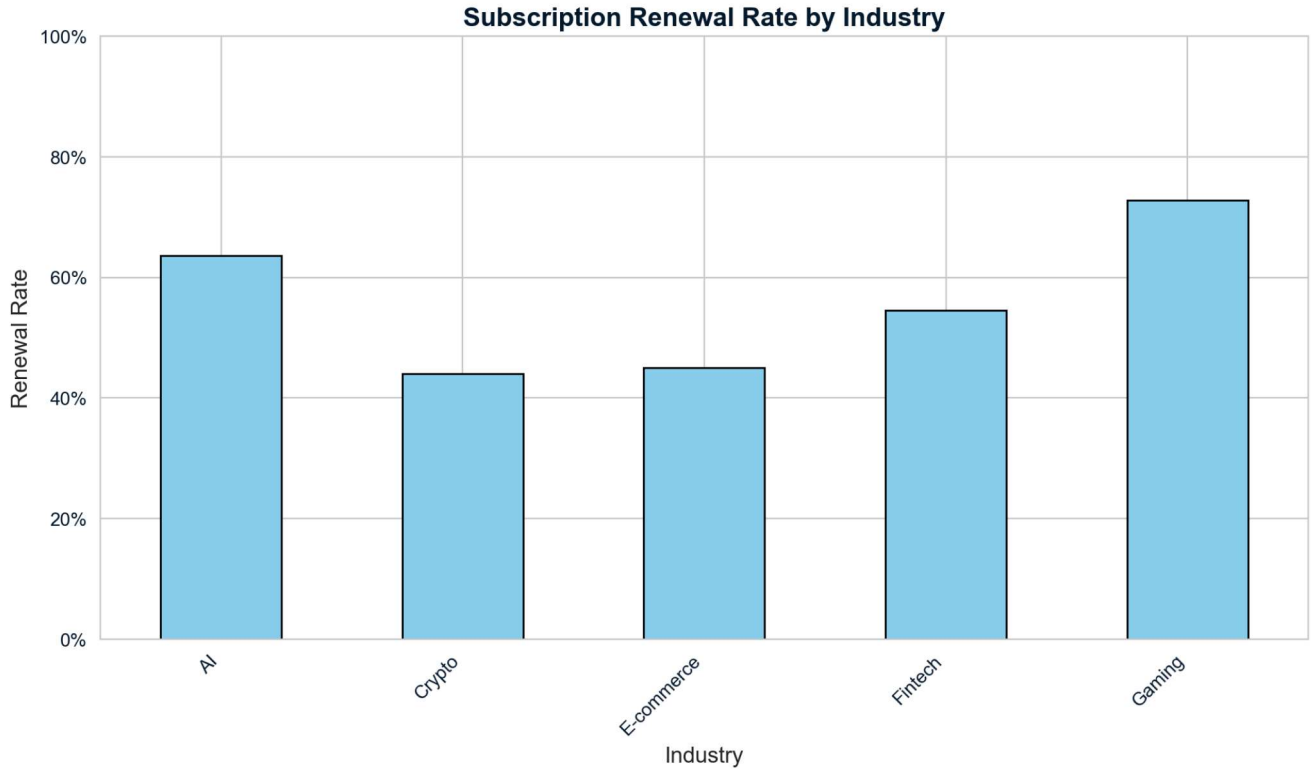
```
# Start coding here, good luck!
##### Question 1 - How many total Fintech and Crypto clients does the company have?
#####
# Define a function that returns 1 if the input is either 'Fintech' or 'Crypto',
otherwise returning 0
def is_fintech_or_crypto(x):
    if x in ['Fintech','Crypto']:
        return 1
    else:
        return 0

# Loop through the 'industry' column in client_details and increment the
total_fintech_crypto_clients counter for every Fintech or Crpyto client
total_fintech_crypto_clients = 0
for industry in client_details['industry']:
    total_fintech_crypto_clients += is_fintech_or_crypto(industry)

# Alternate approach 1 - Apply the custom function directly to the 'industry' column
to calculate the total number of Fintech and Crypto clients
# total_fintech_crypto_clients =
client_details['industry'].apply(is_fintech_or_crypto).sum()

# Alternate approach 2 - Use a lambda function to calculate the total number of
Fintech and Crypto clients
# total_fintech_crypto_clients = client_details['industry'].apply(lambda x: x in
['Fintech','Crypto']).sum()


##### Question 2 - Which industry has the highest renewal rate?   #####
# Merge client details with subscription records
subscriptions = pd.merge(subscription_records, client_details, on = 'client_id', how
= 'left')

# Group by industry and calculate renewal rate
industry_renewal_rates = subscriptions.groupby('industry')['renewed'].mean()

# Find the industry with the highest renewal rate, save as variable 'top_industry'
top_industry = industry_renewal_rates.sort_values(ascending = False).index[0]


##### Question 3 -For clients that renewed their subscriptions, what was the average
inflation rate when their subscriptions were renewed? #####
# Merge subscription records with economic indicators to get the inflation rate at
the subscription end date (i.e., renewal time)
subscriptions_with_inflation =
pd.merge_asof(subscription_records.sort_values(by='end_date'),
                                              economic_indicators,
```

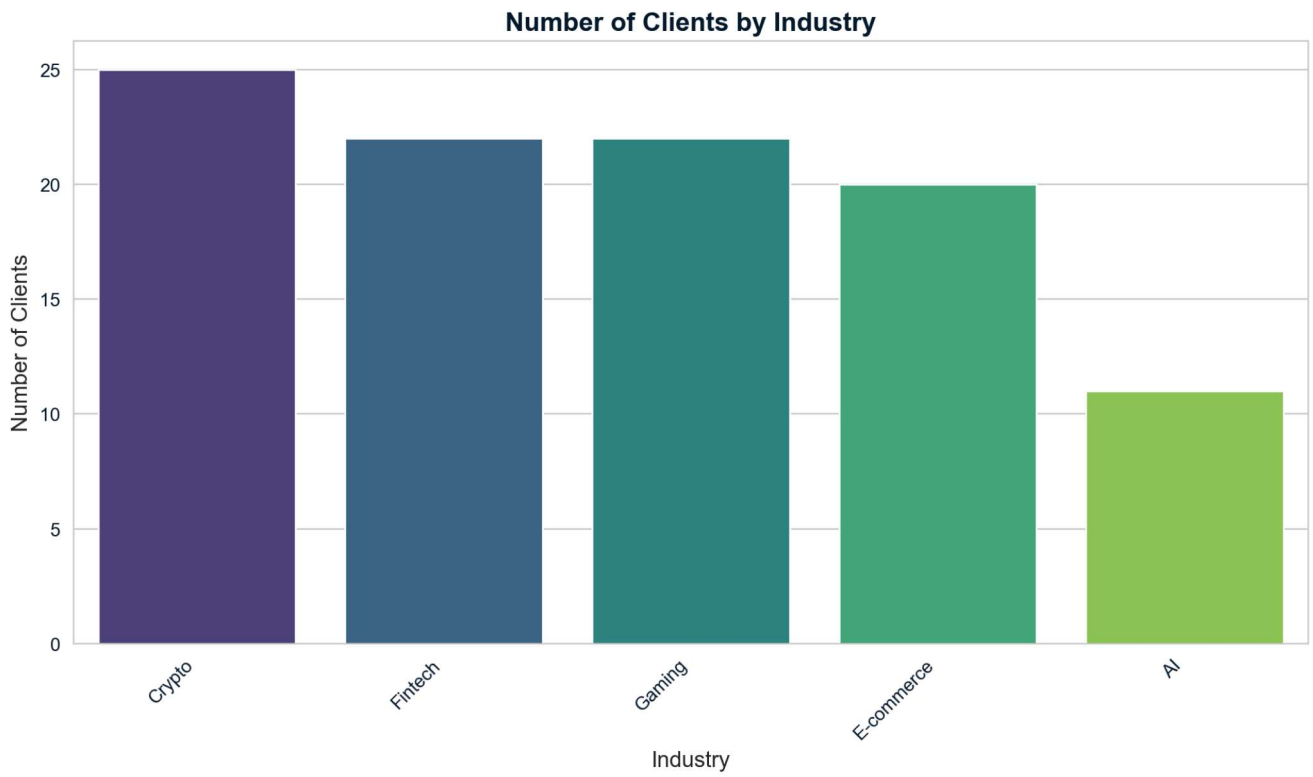
```
                                                left_on='end_date',
                                                right_on='start_date',
                                                direction='backward')

# Calculate the average inflation rate for renewed subscriptions
average_inflation_for_renewals =
subscriptions_with_inflation[subscriptions_with_inflation['renewed'] ==
True].inflation_rate.mean()
```

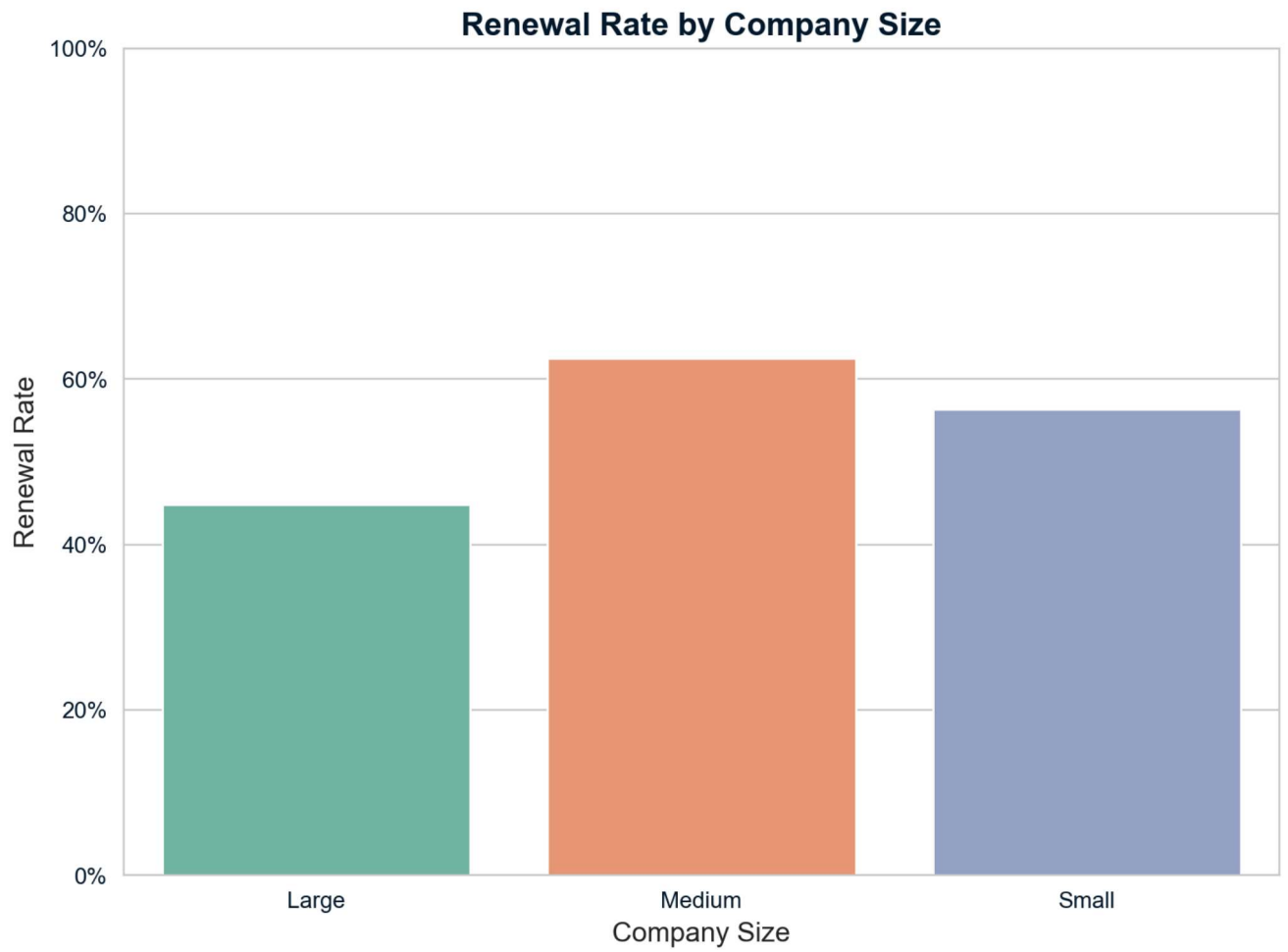
```
# Import visualization libraries
import matplotlib.pyplot as plt
import seaborn as sns

# Set style for better-looking plots
sns.set_style("whitegrid")
plt.rcParams['figure.figsize'] = [10, 6]
plt.rcParams['font.size'] = 10

# ============================================
# Plot 1: Renewal Rate by Industry
# ============================================
plt.figure(figsize=(10, 6))
industry_renewal_rates.plot(kind='bar', color='skyblue', edgecolor='black')
plt.title('Subscription Renewal Rate by Industry', fontsize=14, fontweight='bold')
plt.xlabel('Industry', fontsize=12)
plt.ylabel('Renewal Rate', fontsize=12)
plt.xticks(rotation=45, ha='right')
plt.ylim(0, 1)
plt.gca().yaxis.set_major_formatter(plt.FuncFormatter(lambda y, _: f'{y:.0%}'))
plt.tight_layout()
plt.show()

# ============================================
# Plot 2: Client Count by Industry
# ============================================
plt.figure(figsize=(10, 6))
client_counts =
client_details['industry'].value_counts().sort_values(ascending=False)
sns.barplot(x=client_counts.index, y=client_counts.values, palette='viridis')
plt.title('Number of Clients by Industry', fontsize=14, fontweight='bold')
plt.xlabel('Industry', fontsize=12)
plt.ylabel('Number of Clients', fontsize=12)
plt.xticks(rotation=45, ha='right')
plt.tight_layout()
plt.show()

# ============================================
# Plot 3: Renewal Rate by Company Size
# ============================================
subscriptions_with_clients = pd.merge(subscription_records, client_details,
on='client_id', how='left')
plt.figure(figsize=(8, 6))
size_renewal_rates = subscriptions_with_clients.groupby('company_size')
['renewed'].mean()
sns.barplot(x=size_renewal_rates.index, y=size_renewal_rates.values, palette='Set2')
plt.title('Renewal Rate by Company Size', fontsize=14, fontweight='bold')
```

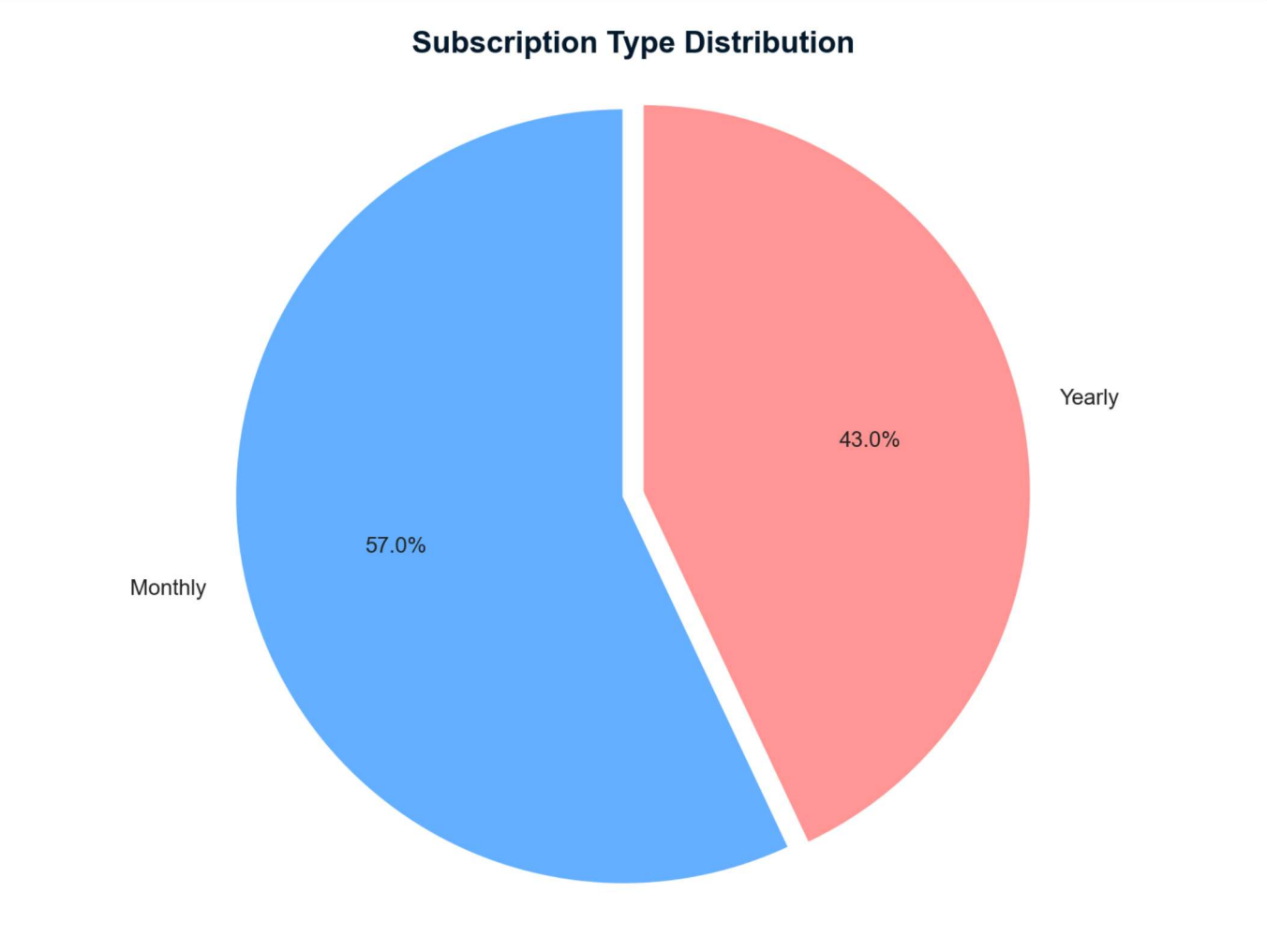
```
plt.xlabel('Company Size', fontsize=12)
plt.ylabel('Renewal Rate', fontsize=12)
plt.ylim(0, 1)
plt.gca().yaxis.set_major_formatter(plt.FuncFormatter(lambda y, _: f'{y:.0%}'))
plt.tight_layout()
plt.show()

# ============================================
# Plot 4: Subscription Type Distribution
# ============================================
plt.figure(figsize=(8, 6))
sub_type_counts = subscription_records['subscription_type'].value_counts()
plt.pie(sub_type_counts.values, labels=sub_type_counts.index, autopct='%1.1f%%',
        colors=['#66b3ff', '#ff9999'], startangle=90, explode=(0.05, 0))
plt.title('Subscription Type Distribution', fontsize=14, fontweight='bold')
plt.axis('equal')
plt.tight_layout()
plt.show()

# ============================================
# Plot 5: Renewals Over Time with Inflation
# ============================================
# Prepare data for time series analysis
renewals_over_time = subscription_records[subscription_records['renewed'] ==
True].copy()
renewals_over_time['renewal_month'] =
renewals_over_time['end_date'].dt.to_period('M')
renewal_counts = renewals_over_time.groupby('renewal_month').size()

# Merge with economic indicators for inflation trend
econ_merged = pd.merge_asof(
    subscription_records.sort_values('end_date'),
    economic_indicators,
    left_on='end_date',
    right_on='start_date',
    direction='backward'
)

econ_merged = econ_merged.reset_index(drop=True)

fig, ax1 = plt.subplots(figsize=(12, 6))
color1 = 'tab:blue'
ax1.set_xlabel('Date', fontsize=12)
ax1.set_ylabel('Number of Renewals', color=color1, fontsize=12)
# Fix: Convert renewal_counts.index to timestamps for plotting
renewal_counts.index = renewal_counts.index.to_timestamp()
renewal_counts.plot(ax=ax1, color=color1, marker='o', label='Renewals')
ax1.tick_params(axis='y', labelcolor=color1)
ax1.grid(True, alpha=0.3)
```

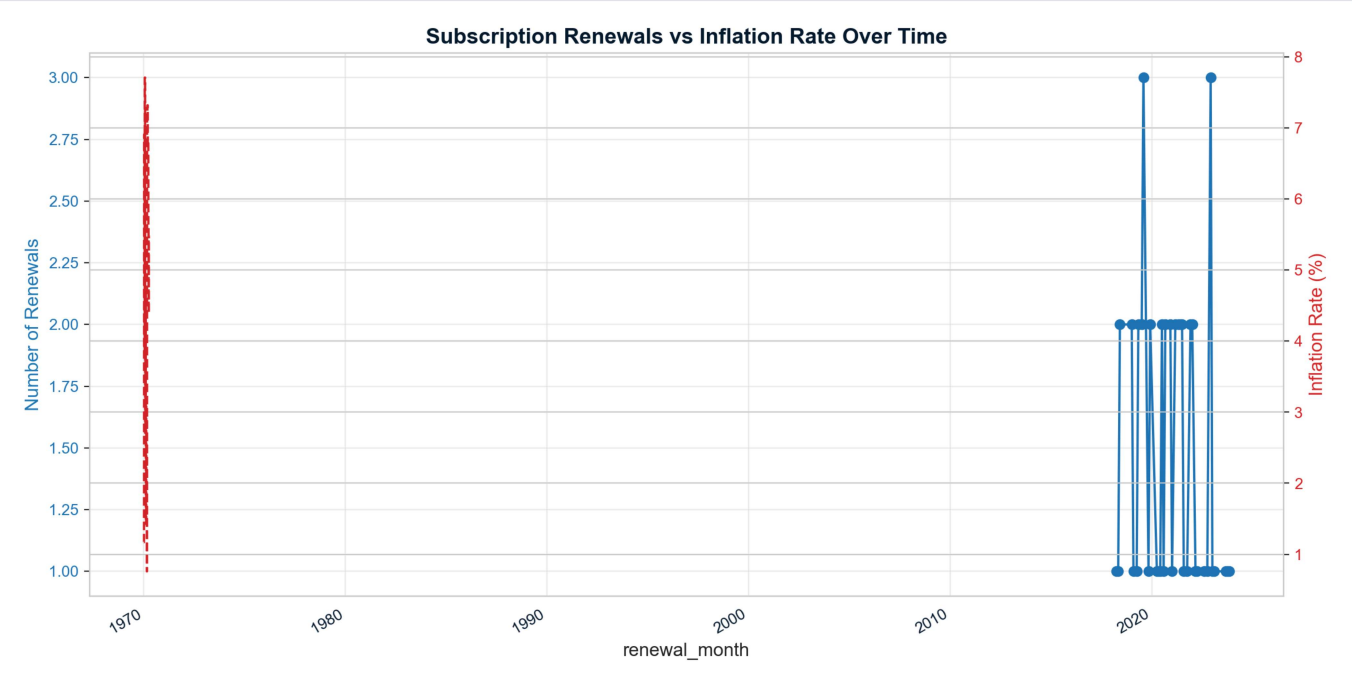
```
ax2 = ax1.twinx()
color2 = 'tab:red'
ax2.set_ylabel('Inflation Rate (%)', color=color2, fontsize=12)
# Fix: Ensure 'end_date' is datetime and use as index for inflation_series
if 'end_date' in econ_merged.columns:
    econ_merged['end_date'] = pd.to_datetime(econ_merged['end_date'])
    inflation_series = econ_merged.groupby('end_date')['inflation_rate'].first()
    inflation_series = inflation_series.sort_index()
else:
    inflation_series = econ_merged['inflation_rate']
# Fix: Plot inflation_series with a supported index (DatetimeIndex)
inflation_series.plot(ax=ax2, color=color2, linestyle='--', label='Inflation')
ax2.tick_params(axis='y', labelcolor=color2)

plt.title('Subscription Renewals vs Inflation Rate Over Time', fontsize=14,
fontweight='bold')
fig.tight_layout()
plt.show()

# ============================================
# Print Key Insights
# ============================================
print(f"✓ Top industry by renewal rate: {top_industry}")
print(f"✓ Average inflation rate at renewal: {average_inflation_for_renewals:.2%}")
print(f"✓ Total Fintech & Crypto clients: {total_fintech_crypto_clients}")
```




Number of Clients by Industry
Number of Clients
25
20
15
10
5
0
Crypto
Fintech
Gaming
E-commerce
AI
Industry


Renewal Rate by Company Size
Renewal Rate
100%
80%
60%
40%
20%
0%
Large
Medium
Small
Company Size






```
✓ Top industry by renewal rate: Gaming
✓ Average inflation rate at renewal: 441.89%
✓ Total Fintech & Crypto clients: 47
```